Development of a Home-scale Prototype to Detect Preservatives in Milk

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2025 IEEE Applied Sensing Conference (APSCON) | 979-8-3503-7933-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/APSCON63569.2025.11144023

***Abstract*—This study presents the design, development, and calibration of a prototype for detecting milk adulteration using an ESP32 microcontroller. The prototype employs a sensor unit to measure the voltage of the milk samples for analysis. Fresh Cow milk was used as unadulterated samples, while sodium hydroxide and hydrogen peroxide were used as adulterants. The calibration process involved using DI water to ensure measurement consistency. Results indicated that including NaOH decreases the impedance (Z) of the sample, leading to higher output voltage, while H_2O_2 increases Z, resulting in lower output voltages. These changes were monitored across different adulteration levels, showing a monotonic relationship between adulterant concentration and output voltage. The prototype demonstrated high repeatability and accuracy, suggesting its potential for home-scale, portable milk adulteration detection.**

***Index Terms*—Milk Preservatives, Impedance, Sensor, Nickle Electrodes**

I. Introduction

Milk is a widely consumed food valued for its nutritional benefits [1]. The high consumer demand for milk as a nutritious food has led to an increase in deceptive adulteration practices aimed at maximizing financial gains [2]. Milk adulteration can occur due to various factors, which can be broadly categorized into four primary motivations: (i) Increasing the volume of liquid in the milk. (ii) Adjusting the levels of fat and non-fat solids. (iii) Falsely elevating the protein content. (iv) Preserving the freshness and quality of the milk [3], [4]. Unfortunately, from a consumer's perspective, there is a lack of accessible and affordable devices to detect preservatives and assess milk quality. In this paper, we developed a prototype that can detect preservatives in milk. A sensor previously developed by this research group has been used here. As reported in our previous work [5], the sensor is a 5 cm high, 1 cm wide, 1 cm long plastic cuvette. We created two identical 6 cm x 1 cm electrodes using 50 µm thick Nickle (Ni) foil. These Ni electrode sheets were bonded to the inner walls of the cuvette using adhesive, forming a parallel capacitor cell within the plastic cuvette. Due to the presence of ions in milk, it exhibits dielectric properties, and its impedance or conductivity depends on its chemical composition [6], [7]. Changes in the milk's composition due to adulteration can be indirectly detected by calculating electrical parameters such as impedance, phase angle, and parallel plate capacitance [5], [8]. Here, impedance in terms of voltage was targeted to identify adulteration in milk. By filling the cuvette with a predetermined amount of milk to maintain a constant effective area "A" and the distance "d" between parallel plates of the sensor, changes in electrical parameters can be attributed to adulteration [5], [9], [10]. This novel design aims to facilitate the rapid detection of milk at consumers' point of use. This prototype also reduces the requirement of milk volume to 4 ml [11]. The present paper discusses the hardware properties, the prototype block diagram, and the measurement results for various samples.

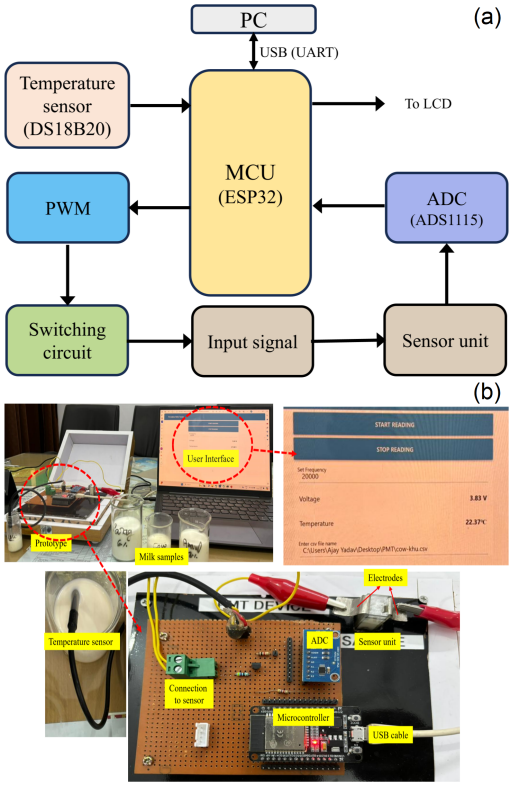

Fig. 1: (a) Block diagram of the prototype (b) Developed prototype

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II. Prototype Development

The circuit of the prototype was designed to function optimally with the fewest number of components needed, making it portable and inexpensive. At the core was the ESP32 microcontroller, known for its high performance and versatility. The Analog Digital Converter (ADC) component was the ADS1115, chosen for its precision in converting analog signals to digital data. Temperature monitoring was handled by the DS18B20 sensor, which provides reliable and accurate temperature readings. Additionally, the BC547 transistor switches the input pulses to the sensor. The block diagram of the prototype is given in Fig. 1 (a). A Novel sensor is utilized here to detect the impedance in terms of the voltage of the various milk samples. When milk samples are adulterated with different adulterants, their chemical composition changes, which in turn affects the impedance or conductivity of the milk. This change in impedance is then converted into the voltage with the help of a load resistance. The final device is being subjected to agile development with a monitoring digital display to be attached. Currently, a laptop or PC can be connected via a USB cable to record and measure parameters. This USB cable also supplies the 5 V power to the ESP32 microcontroller. To power the ADC (ADS1115) and the temperature sensor (DS18B20), different pins on the ESP32 are utilized. Interestingly, the microcontroller also generates a 3.3 V square wave input signal for the sensor electrodes. This means that no additional function generator was required to provide alternating signals of different frequencies to the sensor electrodes. The prototype hardware is housed in a wooden box for safety. All components are soldered and fabricated onto a PCB board. A slot is created in the board itself to securely hold the sensor. The top view of the prototype connected to the laptop is shown in Fig. 1 (b).

III. Methodology

A. Operating Frequency and Voltage

The user interface for operating the prototype was created using Node-RED, allowing the selection of desired frequencies via the dashboard and recording the output voltage and temperature of the sample. In previous work, this research group measured Z, θ, and C_p over a frequency range from 100 Hz to 1 MHz, discovering that Z is particularly frequency-independent above 10 KHz for milk samples [5], [12]. Consequently, 20 kHz frequency was chosen to measure the output voltage of the samples. Regarding the input signal to the sensor electrodes, a constant voltage of 3.3 V was applied to one electrode through the load resistor, while the second electrode was connected to the PWM pin of the ESP32 through a transistor. The transistor is configured to activate and complete the circuit between the two electrodes whenever a positive pulse is received from the PWM pin of the ESP32. Regarding the process of obtaining the output voltage from the current passing through the electrodes, we use a current-to-voltage conversion technique. When a liquid sample, such as milk, is present between the electrodes, the current flows through both the sensor and the liquid. The chemical composition of the liquid determines its impedance

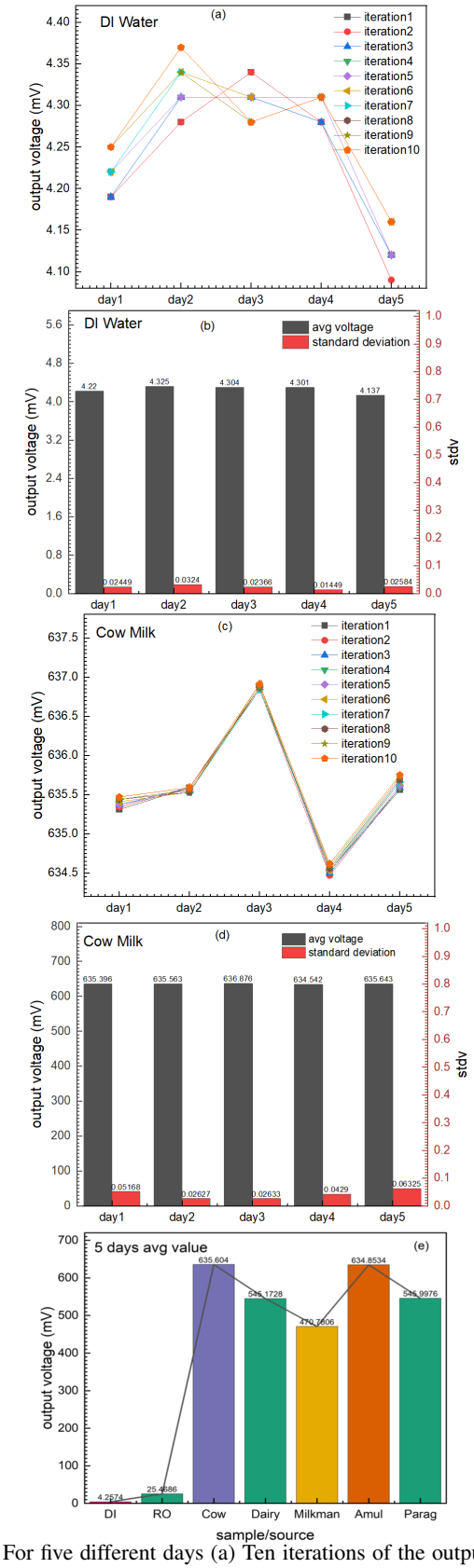

Fig. 2: For five different days (a) Ten iterations of the output voltage for DI water (b) Average voltage plot with standard deviation for DI water (c) Ten iterations of the output voltage for Cow milk (d) Average voltage plot with standard deviation for Cow milk (e) Voltage response of different samples

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(or conductivity), which in turn affects the magnitude of the current flowing through the circuit. As this current passes through the load resistor, a voltage drop is generated across it, which directly corresponds to the composition of the sample.

B. Sample Preparation

The samples were prepared using cow milk collected from a dairy managed by the NGO Khushhali Foundation in Bareilly, U.P., India. We observed the milking process firsthand, which assured us of the milk's purity and lack of adulteration, so it is termed unadulterated samples (US). Two commonly used preservatives, NaOH and H_2O_2, were used to prepare adulterated milk samples. The chemicals were procured from Thermo Fisher India Pvt Ltd. 100 ml of adulterated milk samples were prepared by volumetrically combining NaOH and H_2O_2 as per the adulteration concentration like for 0.5 % adulteration 0.5 ml adulterant was mixed in 99.5 ml milk. In this study, samples adulterated with NaOH are referred to as ASS, while those adulterated with H_2O_2 are designated as ASH. Finally, 4 ml of the adulterated samples were utilized to fill the sensor to determine the output voltage. After each measurement, the sensor unit is cleaned with water and cotton buds and then dried to prepare it for the next measurement.

C. Calibration

The calibration process for this prototype is straightforward. Before measuring the milk samples, a DI water sample is checked to ensure consistency. This verified that the voltage of the DI water matches previous measurements. As shown in Fig. 2 (a), the output voltage for DI water remains within a similar range over five days, ranging from 4.12 mV to 4.32 mV. Each day, ten iterations were recorded to assess repeatability. Figure 2 (b) displays the daily averages of these iterations along with their standard deviations (stdv), which range from 0.0144 to 0.0324. This indicates that variations in the magnitude of output voltages are negligible. Therefore, measurements begin with DI water each day to verify the sensor's repeatability.

IV. Results and Discussion

Since this sensor was designed to produce stable results at a constant temperature range (25 to 30 ^{o}C) [12]. Therefore, all the measurements were performed within this given range. The operating frequency for the measurements was also kept constant at 20 kHz. The responses of the samples were measured in terms of voltage for recording and displaying the results.

A. Voltage Response for Different Milk Samples

For each measurement, 4 ml of the cow milk samples are filled into the sensor unit using a micropipette to maintain volumetric consistency. The prototype was programmed to start the measurement using the dashboard, and the measurement could be stopped at any time via the dashboard itshelf. Ten iterations were taken for each sample measurement. For US milk, Fig. 2 (c) and (d) show the 10 iterations of output voltage and the average output voltage values with standard deviations for five days, respectively. These plots indicate

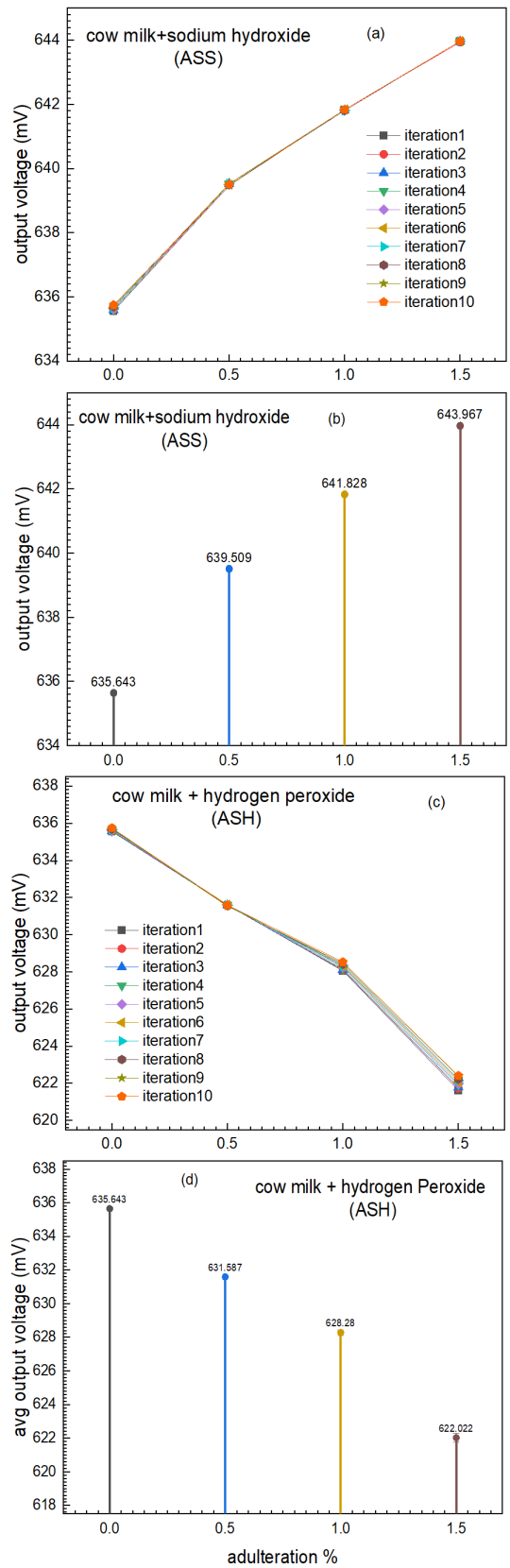

Fig. 3: Voltage versus adulteration, for sodium hydroxide adulterated milk (a) Ten iterations of the output voltage (b) Average output voltage, for hydrogen peroxide adulterated milk (c) Ten iterations of the output voltage (d) Average output voltage

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that the output voltages for US milk samples range from 633 mV to 636 mV, with standard deviations ranging from 0.0126 to 0.115, suggesting that the prototype consistently measures output voltages for milk from the same source over different days. As shown in Fig. 2 (e), the voltage responses for DI and RO water are clearly distinct from those of the milk samples. Noticeable shifts in the voltage responses among the milk samples, likely due to their varying chemical compositions or unknown adulteration, highlight the influence of source-specific properties. To address this variability, our prototype will incorporate an LCD interface that allows users to select the type of milk source to accommodate source-related variations.

B. Voltage Response for ASS

ASS samples were introduced into the sensor to measure the output voltage across the sensor electrodes. These experiments were conducted over five days, similar to the procedure for US milk. Due to the change in salt concentration in these samples, the Z or conductivity was altered. The addition of sodium hydroxide decreases the Z of the sample, thereby increasing the magnitude of the output voltage compared to the unadulterated sample. Figure 3 (a) presents the output voltages of ten iterations from a particular day, showing the voltages of ten iterations almost overlapping. Figure 3 (b) plots the average magnitude of voltage against adulteration concentrations of 0.5 %, 1 %, and 1.5 %. At 0.5 % adulteration, the output voltage changes from 635.6 mV to 639.8 mV. At 1 % adulteration, the output voltage increases to 641.8 mV, and at 1.5 % adulteration, it further rises to 643.9 mV. This monotonic increase in voltage with higher adulteration concentrations facilitates the indirect detection of adulterants, making the process much easier.

C. Voltage Response for ASH

The chemical alteration caused by H_2O_2 adulteration resulted in an increase in the magnitude of Z, leading to a decrease in the output voltage across the sensor electrodes compared to the unadulterated milk samples. This was contrary to the response observed with NaOH inclusion in the milk samples. The inclusion of NaOH increased the salt concentration of the sample, reducing Z or enhancing conductivity, whereas the mixing of H_2O_2 reduced the conductivity of the milk samples due to the presence of excessive H^+ ions [13], and this reduced the output voltage. Ten iterations of output voltages and their average with respect to adulteration concentration are plotted in Fig. 3 (c) and 3 (d), respectively. According to Figure 3 (d), the output voltage for 0.5 % H_2O_2 adulteration decreases from 635.6 mV to 631.5 mV. At 1 % adulteration, it further decreases to 628.2 mV, and at 1.5 % adulteration, it drops to 622.02 mV. Similar to NaOH, the trend in output voltage is monotonic and linear but in the opposite direction.

V. Conclusion

The developed prototype effectively detected milk adulteration by measuring changes in impedance converted to voltage. The calibration process using DI water ensures consistent and reliable measurements, establishing a baseline for comparison with adulterated samples. Our findings demonstrate that the inclusion of NaOH in milk increases its conductivity, resulting in higher output voltages, whereas H_2O_2 decreases conductivity, leading to lower output voltages. This opposite behavior highlights the distinctive effects of different adulterants on milk properties. The study observed a clear monotonic relationship between adulterant concentration and output voltage. Specifically, the output voltage increased with higher concentrations of NaOH and decreased with higher concentrations of H_2O_2. This predictable trend simplifies the identification and quantification of adulterants, enhancing the prototype's practical utility. Throughout the experiments, the prototype exhibited high repeatability, with minimal variation in output voltages across multiple iterations and different days. The standard deviation of the measurements remained low, further affirming the reliability of the prototype. This consistency is crucial for a diagnostic tool intended for widespread use. Future work includes integrating a digital display for real-time monitoring and further automating the measurement process.

References

[1] N. W. Smith, A. J. Fletcher, J. P. Hill, and W. C. McNabb, "Modeling the contribution of milk to global nutrition," Frontiers in Nutrition, vol. 8, p. 716100, 2022.

[2] M. Guetouache, B. Guessas, and S. Medjekal, "Composition and nutritional value of raw milk," J Issues Biol Sci Pharm Res, vol. 2350, p. 1588, 2014.

[3] K. Sharma and M. Paradkar, "Erratum to: The melamine adulteration scandal," Food Security, vol. 2, no. 1, p. 109, 2010.

[4] A. K. Yadav, M. Gattupalli, K. Dashora, and V. Kumar, "Key milk adulterants in india and their detection techniques: a review," Food Analytical Methods, vol. 16, no. 3, pp. 499–514, 2023.

[5] A. K. Yadav, V. Kumar, R. Dhaka, and K. Dashora, "Innovative sensor design for reliable detection of formaldehyde adulteration in milk," Sensors and Actuators A: Physical, p. 115448, 2024.

[6] S. N. Jha, K. Narsaiah, A. L. Basediya, R. Sharma, P. Jaiswal, R. Kumar, and R. Bhardwaj, "Measurement techniques and application of electrical properties for nondestructive quality evaluation of foods—a review," Journal of food science and technology, vol. 48, pp. 387–411, 2011.

[7] S. Das, M. Sivaramakrishna, K. Biswas, and B. Goswami, "A low cost instrumentation system to analyze different types of milk adulteration," ISA transactions, vol. 56, pp. 268–275, 2015.

[8] A. Afzal, M. Mahmood, I. Hussain, and M. Akhtar, "Adulteration and microbiological quality of milk (a review)," Pakistan Journal of Nutrition, vol. 10, no. 12, pp. 1195–1202, 2011.

[9] G. Durante, W. Becari, F. A. Lima, and H. E. Peres, "Electrical impedance sensor for real-time detection of bovine milk adulteration," IEEE Sensors Journal, vol. 16, no. 4, pp. 861–865, 2015.

[10] S. Das, M. Sivaramakrishna, K. Biswas, and B. Goswami, "Performance study of a 'constant phase angle based'impedance sensor to detect milk adulteration," Sensors and actuators A: Physical, vol. 167, no. 2, pp. 273–278, 2011.

[11] M. Chakraborty and K. Biswas, "Limit of detection for five common adulterants in milk: A study with different fat percent," IEEE Sensors Journal, vol. 18, no. 6, pp. 2395–2403, 2018.

[12] A. K. Yadav, K. Dashora, V. Kumar, and R. Dhaka, "Evaluating impedance and phase angle for milk formaldehyde detection," in 2024 IEEE Applied Sensing Conference (APSCON), 2024, pp. 1–4.

[13] S. Biswas, M. Chakraborty, and K. Biswas, "Detection of formaldehyde by a rgo/pmma coated sensor," in 2020 IEEE International Instrumentation and Measurement Technology Conference (I2MTC). IEEE, 2020, pp. 1–6.

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